1.What is Plotly?

Plotly is an open-source Python visualization library used to create interactive, web-based charts and dashboards.

Features:

• Interactive (zoom, hover, filter)

• Supports 2D & 3D plots

• Works in Jupyter Notebook and web apps

• Integrates with Dash for dashboards

It is widely used in data analysis, business intelligence, and reporting applications.

2.What are the different interfaces available in Plotly?

Plotly provides three main interfaces:

1. Plotly Express (px)

o High-level interface

o Simple syntax

o Quick visualizations

2. Graph Objects (go)

o Low-level interface

o Full customization

o Complex visualizations

3. Dash Framework

o Used to build interactive web dashboards

3. Difference between Plotly Express and Graph Objects

Plotly Express

High-level API

Graph Objects

Less code

Low-level API

More control

Best for quick plots

Automatically handles legends & layout Manual configuration required

Best for customized plots

Example:

Plotly Express:

```
import plotly.express as px

fig = px.line(df, x="Date", y="Sales")

fig.show()
```

Graph Objects:

```
import plotly.graph_objects as go

fig = go.Figure()

fig.add_trace(go.Scatter(x=df["Date"], y=df["Sales"]))

fig.show()
```

4.How do we create subplots in Plotly?

Use make_subplots() from plotly.subplots.

```
from plotly.subplots import make_subplots

import plotly.graph_objects as go

fig = make_subplots(rows=1, cols=2)

fig.add_trace(go.Bar(y=[2,3,1]), row=1, col=1)

fig.add_trace(go.Scatter(y=[1,2,3]), row=1, col=2)

fig.show()
```

This allows multiple plots in a single figure.

5.How do you export Plotly charts?

Plotly charts can be exported as:

• HTML file

• PNG

• JPG

• PDF

• SVG

Example:

```
fig.write_html("chart.html")

fig.write_image("chart.png")
```

Note: kaleido package is required for static image export.

6. How to make interactive dashboards?

Interactive dashboards can be created using:

• Dash (Plotly framework)

• Streamlit

• Panel

Using Dash:

1. Import Dash

2. Create layout

3. Add graphs

4. Define callbacks

5. Run server

7.What is a Dashboard?

A dashboard is a visual interface that displays multiple charts, KPIs, and metrics in a single view for monitoring and decision-making.

It provides:

• Real-time insights

• Interactive filtering

• Comparative analysis

Used in:

• Business analytics

• Healthcare

• Banking

• Sales monitoring

8.How to create a Dashboard? (Using Dash)

Basic Steps:

```
from dash import Dash, html, dcc

import plotly.express as px

app = Dash(__name__)

fig = px.bar(x=["A","B"], y=[10,20])

app.layout = html.Div([

html.H1("Sales Dashboard"),

dcc.Graph(figure=fig)

])

if __name__ == "__main__":

app.run_server(debug=True)
```

Steps:

1. Create Dash app

2. Define layout using HTML components

3. Add charts using dcc.Graph

4. Run server

9.What is the use of Heatmap?

A heatmap is used to visualize matrix-form data using colors.

Uses:

• Correlation matrix visualization

• Feature importance comparison

• Performance metrics

• Confusion matrix

Example:

px.imshow(correlation_matrix)

It helps identify strong relationships quickly.

10.What are the different types of Dashboards?

1. Operational Dashboard

o Real-time monitoring

o Example: Server performance

2. Strategic Dashboard

o Long-term decision making

o Example: Annual sales growth

3. Analytical Dashboard

o Detailed data analysis

o Used by data analysts

4. Tactical Dashboard

o Department-level performance

o Example: Marketing campaign performance